Chapter 1. Generating Data

Sampling, experiments, randomization and blocking

Contents

1.1 What is statistics? **2**

1.2 Simple random sampling **2**

1.3 Cautions in sample surveys **3**

1.4 Statistical experiments **4**

1.5 Comparative experiments and randomization **5**

1.6 Blocking **6**

1.7 Selected exercises **7**

1.1 What is statistics?

Motivation. Statistics draws conclusions about a large group from a small part of it, with assessable reliability.

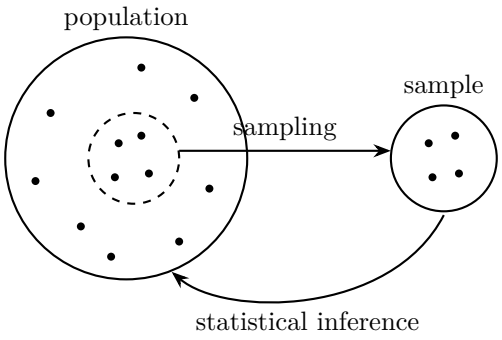
Table 1-1 shows a poll taken before a mayoral election.

Candidate	Support
John Smith	42%
Emily Johnson	37%
Michael Lee	15%
Sarah Davis	6%

Three tasks arise: *whom* to survey (how many, chosen how, so as to represent all voters); how to *summarize* the data and conclude about all voters (does Smith lead Johnson overall?); and how to *assess the reliability* of the conclusion. **Statistics** is the science that collects, organizes and interprets numerical information (**data**) to reach rational conclusions with a measurable degree of reliability. Since surveying every unit is usually too costly or impossible, we observe a part and reason about the whole.

Definition (Population and sample)

Each individual unit making up the whole is a **sampling unit**; the value describing the feature of interest of a unit (support for a candidate, sleep quality, drug effect) is a **characteristic**. The collection of the characteristic values of *all* sampling units is the **population**; the collection of values actually observed is the **sample**.
A population made of finitely many units is a **finite population** (all voters of a city). A population with infinitely many units, e.g. all present and future smartphone users, is an **infinite population**; it is a conceptual, not a physically existing, object.



A brief history. "Statistics" comes from *State Arithmetics* (censuses, land valuation for taxes). In the 17th–18th centuries astronomers and surveyors met **variability** of repeated measurements; in the 19th century statistics spread to agriculture and biology, and in the 20th it became a unified science, now used from epidemic forecasting and polling to AI evaluation and climate research.

1.2 Simple random sampling

Motivation. Inference from a part is trustworthy only if chance, not people, picks the part.

A satisfaction survey of all Korean students that covers only students near the surveyor's home is not representative.

Illustration 1-1: a biased product sample

To test the strength of steel plates, an inspector cuts ten $5\,\mathrm{cm}^2$ pieces but avoids visible defects. The test says the plates are excellent, though the sample does not reflect actual quality.

Illustration 1-2: voluntary response

Of 5 million viewers of a singing contest, 10% (500,000) voted by web, text or phone; contestant A won with 50%. A random survey of 500 viewers found only 10% preferred A: the voters were mostly A's fans. A sample of self-selected respondents is a **voluntary response sample**; it can badly distort opinion.

Giving every unit an equal chance of selection removes the surveyor's intervention (1-1) and self-selection (1-2), the sources of structural error; this requires careful *sample design*.

Definition 1-1 (Simple random sampling)

A sampling method designed so that, from a finite population, every subset consisting of n sampling units has the same probability of being selected as the sample is called **simple random sampling**.

With population size N, each of the $\binom{N}{n}$ subsets has probability $1/\binom{N}{n}$. Sequentially: the first unit is picked with probability $1/N$, and after k units one of the remaining $N-k$ with probability $1/(N-k)$ (**sampling without replacement**, probabilities $\frac{1}{N}, \frac{1}{N-1}, \ldots, \frac{1}{N-n+1}$). In **sampling with replacement** the chosen unit is returned, so each draw picks one of N units with probability $1/N$.

Proposition (Properties of simple random sampling)

Simple random sampling (without replacement) of size n from a population of size N satisfies
(1) every possible sample of size n has the same chance of being selected;
(2) every unit of the population has the same chance of being selected, namely n/N.

Proof. [Supplement]

Step 1. An ordered list of n distinct units is drawn with probability $\frac{1}{N}\cdots\frac{1}{N-n+1}$; a subset arises from $n!$ orderings, so

$$\frac{n!}{N(N-1)\cdots(N-n+1)} = \frac{n!\,(N-n)!}{N!} = \frac{1}{\binom{N}{n}}$$

for every subset; this is (1).

Step 2. A fixed unit lies in $\binom{N-1}{n-1}$ of the $\binom{N}{n}$ subsets: probability $\binom{N-1}{n-1}/\binom{N}{n} = n/N$. ∎

Systematic sampling (random start, then every k-th unit) satisfies (2) but not (1). Statistical designs use **probability sampling**, with explicitly defined selection probabilities (Chapter 12).

1.3 Cautions in sample surveys

Motivation. Random selection removes the surveyor's bias, but practical errors remain.

(1) Incomplete frame. Units missing from the list (homeless people, households without phones) cause **bias** if they differ from the rest.

(2) Non-response. Non-response often reaches 30%; replacing non-respondents does not help if their views differ.

(3) Response and interviewer effects. Sensitive questions ("Have you ever downloaded illegally?") are answered untruthfully; tone or leading phrasing ("Most people exercise three times a week; how often do you?") biases answers. Interviewers must be trained.

(4) Question wording. "Environmental regulation places an excessive burden on small firms, costing jobs and hindering growth. Do you favour relaxing it?" induces support for deregulation.

1.4 Statistical experiments

Motivation. Surveys observe units as they are; experiments act on them and watch the response.

Definition 1-2 (Experimental unit and treatment)

The individual on which an experiment is performed is an **experimental unit**; applying a specific experimental environment or condition to an experimental unit is a **treatment**.

Definition 1-3 (Response variable, factor, factor level)

In a statistical experiment, a variable expressing the experimental environment or condition is a **factor**, and the variable expressing the reaction to it is the **response variable**. The values a factor takes are the **levels** of the factor.

Illustration 1-3: vaccine dose

One vaccine dose is given to 25 people; the response is blood concentration after 30 minutes. One factor (vaccine) at one level; three doses would give three levels.

Example 1-1 (Diabetes treatment)

Half of 18,000 diabetic patients take a new treatment daily, the other half the existing one; after six months the new-treatment group has lower mean blood sugar. Identify the experimental units, factor, levels and response variable.
Solution. Units: the 18,000 patients. Factor: type of treatment, with two levels (new, existing). Response: blood sugar after six months.

Illustration 1-4: two factors

Do practice problems work best at the start or end of a chapter, depending on problem type? Six textbook versions are used, with equal study time, homework and final test.

Position \ Type	Simple fact	Computation	Concept
End of chapter	Group 1	Group 2	Group 3
Start of chapter	Group 4	Group 5	Group 6

Units: students; response: test score; two factors, position (2 levels) and type (3 levels), whose $2 \times 3 = 6$ combinations are the treatments.

1.5 Comparative experiments and randomization

Motivation. Control groups and random assignment separate a treatment's effect from everything else.

Illustration 1-5: gastric freezing

Gastric freezing (a swallowed balloon with circulating coolant) was claimed to relieve ulcer pain, based on

$$\text{observe pain} \;\rightarrow\; \text{gastric freezing} \;\rightarrow\; \text{observe pain}.$$

Patients improved, but through the **placebo effect**, a positive response to any treatment, which affects even physiological responses such as acid secretion. A controlled study of 160 patients found 34% of 82 frozen and 38% of 78 given saline improved.

Both groups of the second study received the placebo influence, so their difference is the effect of freezing itself, which was nil. The group receiving the treatment is the **treatment group**; the group receiving the dummy treatment is the **control group**.

First principle of experimental design

Minimize the effect of extraneous factors other than the factor of interest.

Neither patient nor experimenter should know the assignment (**double-blind experiment**), and the groups must be alike in extraneous factors; otherwise the group difference mixes the treatment effect with that factor: **confounding**.

Illustration 1-6: confounded study

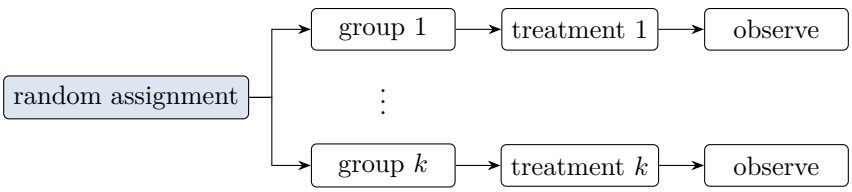
After a chemical leak, an agency reported genetic damage in residents, but age and living habits were not controlled, so the damage could not be attributed to the chemicals. A similar control group (age, sex, region), identical measurement and blind assessment were needed.

Extraneous factors are too many to equalize by hand, and assignment by judgement (e.g. the new drug to milder cases) can bias the result; units should be assigned by chance.

Illustration 1-7: random assignment

Of 30 newly weaned rats, 10 chosen by simple random sampling get a new food and 20 (control) the existing food; weight gain after 4 weeks is measured.

Each rat is treated with probability 1/3. Assigning units by chance so that each has the same chance of each treatment is **randomization** (R. A. Fisher, Rothamsted).



Completely randomized design (Figures 1-2 and 1-3; $k = 2$ is the randomized comparative experiment)

Second principle of experimental design (randomization)

When deciding which treatment each experimental unit receives, give every unit the same chance of receiving any particular treatment.

Subjective assignment produces unexpected errors, while random results obey the laws of probability. "Random" is not "haphazard": taking the first ten rats caught may select smaller or slower ones. For k treatments, draw groups successively by simple random sampling (e.g. 30 units into three groups of 10); this is a **completely randomized design**.

A response difference is still not automatically a treatment effect (fast growers may cluster by chance); randomization lets probability judge whether it is too large for chance (Chapter 6), and more units make groups more alike.

Third principle of experimental design (replication)

Apply each treatment repeatedly to a sufficiently large number of experimental units.

1.6 Blocking

Motivation. Grouping similar units and randomizing within groups gives more precise comparisons.

Illustration 1-8: paired design

Does room temperature (15 °C vs. 20 °C) affect manual dexterity (pegs inserted in 30 minutes)? Individual dexterity varies so much that two separate groups may hide a small effect. Instead every subject works at both temperatures on different days, in random order (heads $\Rightarrow$ 15 °C first).

This **randomized paired comparison design** keeps comparison, randomization and replication; each unit is its own control, removing between-person variation. Pairing two similar subjects and randomizing who works at 15 °C is also a paired design.

Definition (Blocking)

Blocking means forming, before the experiment, groups of experimental units expected to respond similarly to the same treatment; each such group is a **block**. A design that uses blocking and randomization (within blocks) together is a **randomized block design**; the randomized paired comparison design is a special case (block size 2).

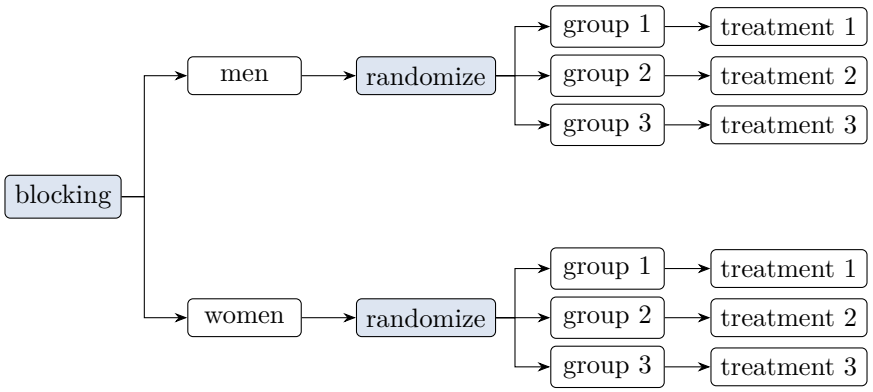
Blocks are built from factors expected to affect the response. Usually block size equals the number of treatments (1-9); larger blocks are split by randomization into treatment groups (1-10).

Illustration 1-9: field blocks

For 2 soybean varieties $\times$ 3 pesticides, each field (soil, fertility) is a block of 6 plots, with the 6 treatments randomized to plots.

Illustration 1-10: blocking by sex

A cancer progresses differently in men and women. Three treatments are tested on 30 male and 30 female patients as two blocks; within each block randomization forms three groups of 10, and survival is compared within blocks.



Randomized block design of Illustration 1-10 (Figure 1-4)

Survey or experimental data are analysed as a *sample from a population*, often a conceptual infinite one: in 1-9, each variety–pesticide combination defines an infinite population of yields.

In an *experiment* the experimenter decides who is treated, so extraneous factors can be controlled; in a *sample survey* units decide themselves. When experiments are unethical (forcing people to smoke), an **observational study** is needed, and hidden extraneous factors require special care.

1.7 Selected exercises

Motivation. Systematic sampling, confounding and blocking, and how randomization balances extraneous factors.

Exercise 1.5 (Systematic sampling)

In systematic sampling the units are listed and selected at a fixed interval. For instance, to choose 4 addresses from a list of 100, the interval is $100/4 = 25$; a number r is drawn at random from $1, \ldots, 25$, and the sample is the addresses numbered r, $r+25$, $r+50$, $r+75$ (for $r = 13$: $13, 38, 63, 88$).
(a) Describe how to select a systematic sample of size 5 from a list of 200 addresses.
(b) Show that every address has the same chance of being selected, but that the method is not simple random sampling.

Solution.

(a) $k = 200/5 = 40$: draw r uniformly from $\{1, \ldots, 40\}$ and take $r,\ r+40,\ r+80,\ r+120,\ r+160$.

(b)

Step 1. The 40 samples $S_r = \{r, r+40, \ldots, r+160\}$ partition the 200 addresses, so $P(j \text{ selected}) = 1/40 = n/N$, as in property (2) (Section 1.2).

Step 2. Only the 40 sets S_r have positive probability, not all $\binom{200}{5} \approx 2.5 \times 10^9$ subsets (addresses $1, 2$ never occur together), so property (1) fails.

If the list has a period matching the interval (every 40th address a corner flat), the sample is all one type and biased. ■

Exercise 1.9 (Fungus and wind)

An experiment compares whether fungal disease spreads differently among three crops A, B and C. Fungal spread is known to be affected by the wind direction. The field is divided into three adjacent strips in a row along the wind direction, and A, B and C are planted in the

three strips, C being nearest to the side from which the wind blows. Point out the problem with this experiment and propose a suitable design.

Solution.

Step 1. (The problem.) Crop and position along the wind are completely *confounded* (C always nearest the wind), and with one plot per crop there is no replication.

Step 2. (A design.) Form *blocks* as bands across the wind, so plots in a block are equally exposed; randomize A, B, C to the three plots of each block independently.

Step 3. (Why it works.) In this randomized block design (Section 1.6) every crop meets every exposure level, within-block comparisons remove exposure differences, randomization guards against unknown plot differences, and the blocks give replication. ■

Exercise 1.11 (A confounded comparison)

To test whether a new learning programme is more effective than the current one, a school randomly splits the 40 students of one class into two groups of 20. One group studies with the new programme, taught by one teacher; the other group studies with the current programme, taught by another teacher. At the end the new-programme group has improved more, and the school concludes that the new programme is effective. Although the students were randomized, the experiment has a serious defect. Explain why the conclusion cannot be trusted.

Solution.

Step 1. The treatment is "programme with teacher": the programme effect is confounded with the teacher effect. Randomizing students balances students, not teachers.

Step 2. With one teacher per programme there is no replication at the programme level, so no measure of teacher-to-teacher variation.

Step 3. Let each teacher teach both programmes to randomly assigned halves (teachers as blocks), or randomize several classes and teachers to programmes. ■

Exercise 1.13 (Randomization balances a covariate)

There are $2N$ experimental units with values $u_1, \dots, u_{2N}$ of some extraneous variable (age, fertility, ...), with mean $\bar{u}$ and $\sigma^2 = \frac{1}{2N}\sum_k (u_k - \bar{u})^2$. A completely randomized design sends a simple random sample of N units to treatment and the rest to control. Let $\bar{u}_T$ and $\bar{u}_C$ be the mean values of u in the two groups. Show that

$$E(\bar{u}_T - \bar{u}_C) = 0, \qquad \operatorname{Var}(\bar{u}_T - \bar{u}_C) = \frac{4\sigma^2}{2N-1}.$$

Solution [Supplement].

Step 1. Let $\delta_k = 1$ if unit k is treated, else 0, and $d_k = u_k - \bar{u}$, so $\sum_k d_k = 0$, $\sum_k d_k^2 = 2N\sigma^2$. As $\sum_k \delta_k = N$ and $\bar{u}_C = 2\bar{u} - \bar{u}_T$,

$$\bar{u}_T - \bar{u} = \frac{1}{N}\sum_k \delta_k u_k - \bar{u} = \frac{1}{N}\sum_k \delta_k d_k, \qquad \bar{u}_T - \bar{u}_C = 2(\bar{u}_T - \bar{u}).$$

Step 2. $P(\delta_k = 1) = \frac{1}{2}$ (property (2), Section 1.2); a pair lies in $\binom{2N-2}{N-2}$ of the $\binom{2N}{N}$ groups, so for $k \neq l$

$$P(\delta_k = \delta_l = 1) = \frac{N(N-1)}{2N(2N-1)}, \qquad \operatorname{Cov}(\delta_k, \delta_l) = \frac{N-1}{2(2N-1)} - \frac{1}{4} = -\frac{1}{4(2N-1)},$$

and $\operatorname{Var}\delta_k = \frac{1}{4}$.

Step 3. (Mean.) $E\sum_k \delta_k d_k = \frac{1}{2}\sum_k d_k = 0$.

Step 4. (Variance.) With $\sum_{k\neq l} d_k d_l = -\sum_k d_k^2$,

$$\operatorname{Var}\Big(\sum_k \delta_k d_k\Big) = \frac{1}{4}\sum_k d_k^2 + \frac{1}{4(2N-1)}\sum_k d_k^2 = \frac{2N}{4(2N-1)}\cdot 2N\sigma^2 = \frac{N^2\sigma^2}{2N-1}.$$

So $\operatorname{Var}(\bar{u}_T - \bar{u}) = \sigma^2/(2N-1)$ and $\operatorname{Var}(\bar{u}_T - \bar{u}_C) = 4\sigma^2/(2N-1)$.

Randomization balances u *on average*, even unmeasured, with chance imbalance shrinking like $1/N$; blocking on u removes it altogether. ■